Implementation of Hierarchical Clustering

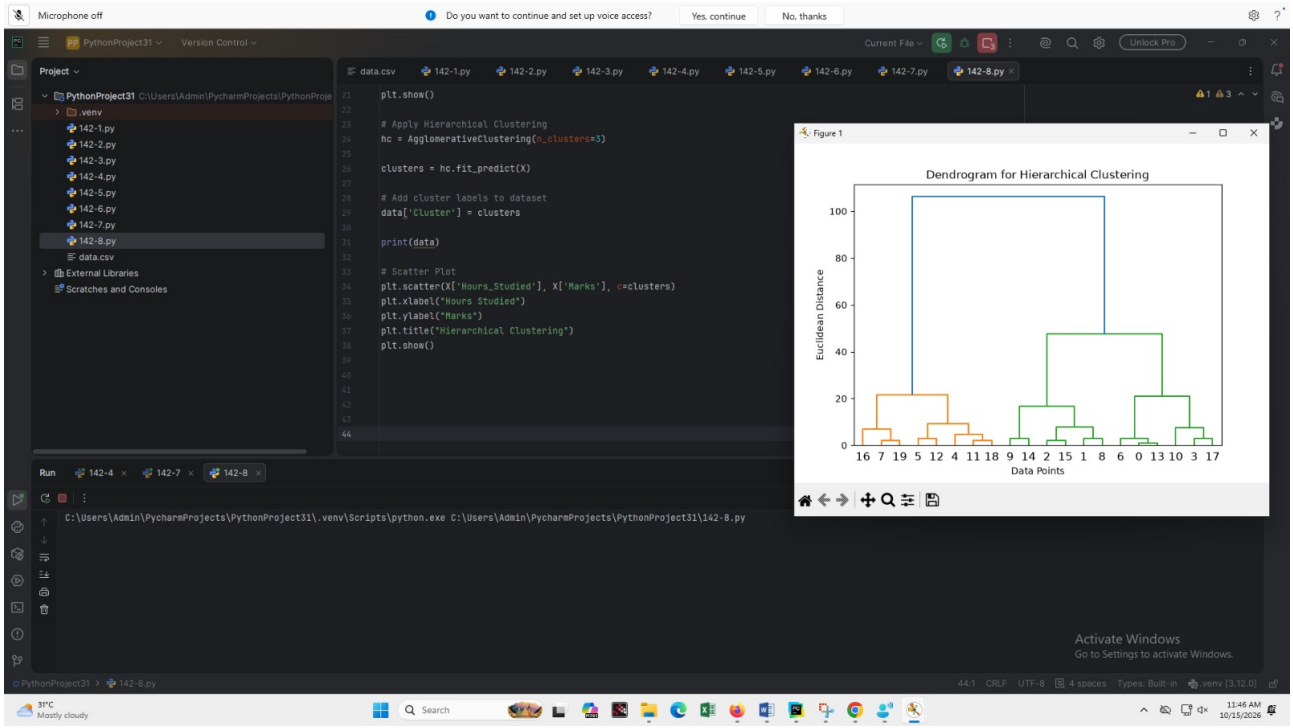
Aim

To write a program that implements hierarchical (agglomerative) clustering to group the given data points into clusters based on their similarity, and to display the cluster labels and a dendrogram showing how the clusters are merged.

Algorithm

1. Start the program.
2. Import the required libraries (pandas, matplotlib, dendrogram and linkage from scipy.cluster.hierarchy, and AgglomerativeClustering from sklearn.cluster).
3. Load the dataset from the input file.
4. Select the input features to be used for clustering (no target variable is needed, since clustering is unsupervised).
5. Treat each data point as its own separate cluster.
6. Compute the distance between every pair of clusters (for example Euclidean distance with Ward linkage).
7. Merge the two closest clusters into a single cluster.
8. Update the distances between the new cluster and all the remaining clusters.
9. Repeat steps 7 and 8 until all the data points are merged into one cluster.
10. Plot the dendrogram to show the merging process and decide the number of clusters by cutting the tree at a suitable height.
11. Create the AgglomerativeClustering model with the chosen number of clusters (for example 3) and fit it on the data to obtain the cluster label for each data point.
12. Display the cluster labels and add them to the dataset.
13. Plot the data points coloured by cluster (optional).
14. Stop the program.

Program


Microphone off
Do you want to continue and set up voice access?
Yes, continue
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PythonProject31
Version Control
Current File
Unlock Pro
Project
PythonProject31 C:\Users\Admin\PycharmProjects\PythonProje
.venv
142-1.py
142-2.py
142-3.py
142-4.py
142-5.py
142-6.py
142-7.py
142-8.py
data.csv
External Libraries
Scratches and Consoles
data.csv
142-1.py
142-2.py
142-3.py
142-4.py
142-5.py
142-6.py
142-7.py
142-8.py
plt.show()
Apply Hierarchical Clustering
hc = AgglomerativeClustering(n_clusters=3)
clusters = hc.fit_predict(X)
Add cluster labels to dataset
data['Cluster'] = clusters
print(data)
Scatter Plot
plt.scatter(X['Hours_Studied'], X['Marks'], c=clusters)
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Hierarchical Clustering")
plt.show()
Figure 1
Dendrogram for Hierarchical Clustering
Euclidean Distance
0
20
40
60
80
100
16 7 19 5 12 4 11 18 9 14 2 15 1 8 6 0 13 10 3 17
Data Points
Run
142-4
142-7
142-8
C:\Users\Admin\PycharmProjects\PythonProject31\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject31\142-8.py
Activate Windows
Go to Settings to activate Windows.
PythonProject31 > 142-8.py
44:1 CRLF UTF-8 4 spaces Types: Built-in .venv (3.12.0)
31°C
Mostly cloudy
Search
11:46 AM
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